Module 11 Assignment

Matrix Theory

Required Problems

Please submit your solutions to these problems to be graded. All solutions should be clearly written and include full explanation/justification.

1. Find a Schur triangularization of the matrix

$$A = \begin{bmatrix} 4 & 4 & 1 \\ -1 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

2. (a) Let $A, B, C \in \mathbb{C}^{n\times n}$. Prove that $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$.

   (b) Prove that if $A \in \mathbb{C}^{n\times n}$ is a matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (not necessarily distinct), then $\operatorname{tr}(A) = \sum_{j=1}^{n} \lambda_j$.

   (c) Prove that if $A \in \mathbb{C}^{n\times n}$ is a matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ (not necessarily distinct), then $\det(A) = \lambda_1\lambda_2\cdots\lambda_n$.

3. (NS 6.5.8) Let T be a linear operator on a complex vector space of dimension 5. Suppose $\chi_T(t) = (t-2)^2(t-3)^3$, $\operatorname{null}(T-2I) = 1$, and $\operatorname{null}(T-3I) = 2$. Find the Jordan normal form of T (up to permutation of the Jordan blocks).



Extra Practice Problems

This is a list of problems I recommend if you would like further practice to help you master the material. You are not expected to solve all of these problems, nor are you required to submit them for grading.

- Section 6.2: 1
- Section 6.3: 2, 4, 7
- Section 6.4: 1, 2, 3, 9, 13, 14
- Section 6.5: 3, 5, 7, 8, 9, 12, 18

1. Prove that if Q is an $n \times n$ unitary matrix, then the block matrix $P = \begin{bmatrix} I_k & 0_{k\times n} \\ 0_{n\times k} & Q \end{bmatrix}$ is a unitary matrix.
2. Prove that if P_1 and P_2 are unitary matrices, then P_1P_2 is a unitary matrix.
3. Verify that the Schur form is not unique by showing that I^*AI and P^*AP are both Schur decompositions of A, where
$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$
4. **Nilpotent Matrices**
   (a) Prove that the Jordan block matrix $J(0, m)$ is nilpotent. (**Hint.** Use induction on k and show that the (i, j)-entry of $J(0, m)^k = 0$ for $i > j - k$.
   (b) Prove that a Jordan matrix is nilpotent if and only if all of the entries on its main diagonal are zero.
   (c) Prove that a matrix is nilpotent if and only if its Jordan normal form is nilpotent.
   (d) Explain why a matrix is nilpotent if and only if its only eigenvalue is 0.



5. **The Cayley-Hamilton Theorem.** Let $A \in \mathbb{C}^{n \times n}$ with characteristic polynomial $\chi_A(t)$. (**Hint.** Throughout this problem, think of the factored form of the characteristic polynomial $(a_n(t-\lambda_1)(t-\lambda_2)\cdots(t-\lambda_n)$.)

   (a) Prove that if D is a diagonal matrix, then $\chi_D(D) = 0$.

   (b) Prove that if A is a diagonalizable matrix, then $\chi_A(A) = 0$.

   (c) Prove that if $J(\lambda, m)$ is a Jordan block, then $\chi_{J(\lambda,m)}(J(\lambda, m)) = 0$.

   (d) Prove that if J is a Jordan matrix, then $\chi_J(J) = 0$.

   (e) Prove the Cayley-Hamilton Theorem: for any square matrix A, $\chi_A(A) = 0$.

